Word–Sentence Framework for Remote Sensing Image Captioning

Qi Wang, *Senior Member, IEEE*, Wei Huang, *Student Member, IEEE*, Xueting Zhang, and Xuelong Li, *Fellow, IEEE*

***Abstract*— Remote sensing image captioning (RSIC), which aims at generating a well-formed sentence for a remote sensing image, has attracted more attention in recent years. The general framework for RSIC is the encoder–decoder architecture containing two submodels of encoder and decoder. Although the significant performance is obtained, the encoder–decoder architecture is a black-box model with a lack of explainability. To overcome this drawback, in this article, we propose a new explainable word–sentence framework for RSIC. The proposed word–sentence framework consists of two parts: word extractor and sentence generator, where the former extracts the valuable words in the given remote sensing image, while the latter organizes these words into a well-formed sentence. The proposed framework decomposes RSIC into a word classification task and a word sorting task, which is more in line with human intuitive understanding. On the basis of the word–sentence framework, some ablation experiments are conducted on the three public RSIC data sets of Sydney-captions, UCM-captions, and RSICD to explore the specific and effective network structures. In order to evaluate the proposed word–sentence framework objectively, we further conduct some comparative experiments on these three data sets and achieve comparable results in comparison with the encoder–decoder-based methods.**

***Index Terms*— Deep learning, image captioning, remote sensing, word–sentence framework.**

I. Introduction

BENEFITTING from the development of remote sensing devices and technologies, many applications of optical remote sensing images have made great progress, such as scene classification [3], [4], disaster detection [5], [6], object detection [7], [8], geographical image retrieval [9], [10], image captioning [11], [12], semantic segmentation [13], [14], and others [15]. However, these tasks mainly explore the visual features and attributes in remote sensing images, such as scene category and object location.

To explore their texture relationship, recently, researchers have studied a novel task of remote sensing image captioning (RSIC) [1], [2], [16], [17], which aims to generate a well-formed sentence to comprehensively and accurately describe the relationship of features and objects in remote sensing images.

In the process of describing a remote sensing image, there are two key problems to be solved: visual feature extraction and texture relationship description. In order to achieve this goal, it is necessary to take advantage of both computer vision (CV) techniques and natural language processing (NLP) techniques, which is also the motivation of natural image captioning (NIC) [18]–[26]. Although the task of RSIC and NIC is the same, there are some differences between remote sensing images and neural images in the following two aspects.

1) There are plenty of multiscale objects and features mixed on the ground in remote sensing images, which means that RSIC needs to pay more attention to understand the whole scene relationship. On the contrary, there is more foreground and background information in neural images.
2) Remote sensing images are only taken from an aerial view, and therefore, the spatial relationship of objects and features in remote sensing images would be simpler and more stable than neural images.

Inspired by NIC, researchers propose many methods in RSIC and achieve significant performance. These methods obey a general framework of encoder–decoder, which consists of encoder submodel and decoder submodel. Encoder submodel extracts visual features from remote sensing images and saves them in a high-dimensional feature vector (or feature map). According to the feature vector, a decoder submodel generates a sentence well-formed in grammar and logic. Although encoder–decoder-based methods achieve excellent performance, there is a disadvantage that the encoder–decoder framework is a black-box model with the lack of explanation of different states during the captioning process. For example, there is no explicit physical meaning for each element in the feature vector. To deal with it, in this article, we propose a novel explainable word–sentence framework for RSIC, which is briefly illustrated in Fig 1.

Similar to the encoder–decoder framework, the proposed word–sentence framework also consists of two submodels: word extractor and sentence generator. Different from the encoder-encoder framework, however, these two submodels play different roles in function. For the word extractor, it takes as input the remote sensing images and as output the valuable but disordered words of all types in the images, including noun, pronoun, numeral, adjective, conjunction, and others. For a sentence generator, it takes as input the disordered

Manuscript received March 27, 2020; revised October 16, 2020; accepted December 8, 2020. Date of publication December 25, 2020; date of current version November 24, 2021. This work was supported by the National Natural Science Foundation of China under Grant U1864204, Grant 61773316, Grant U1801262, and Grant 61871470. *(Corresponding author: Xuelong Li.)*

The authors are with the School of Computer Science, Northwestern Polytechnical University, Xi'an 710072, China, and also with the Center for Optical Imagery Analysis and Learning (OPTIMAL), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: crabwq@gmail.com; hw2hwei@gmail.com; zxt@mail.nwpu.edu.cn; xuelong_li@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2020.3044054

1558-0644 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:26:20 UTC from IEEE Xplore. Restrictions apply.

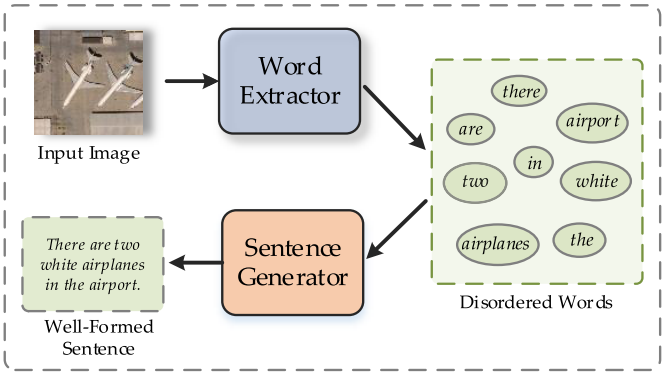


Fig. 1. Illustration of the proposed word–sentence framework for RSIC.

words and as output the ordered sequence of them, i.e., a corresponding sentence well-formed in grammar and logic. To some extent, our framework is more in line with human understanding of image captioning.

In order to meet the requirements of word extraction and sentence sorting, we construct the following two-stage word–sentence framework: word extractor is realized by a CNN-based multilabel classifier where each word is regarded as one category, while sentence generator is realized by a transformer that can deal with the problem of sequence to sequence. To make CNN and transformer more suitable for the task of RSIC, several improvements are made in this article. For the CNN-based multilabel classifier, we attempt four widely used CNNs and optimize them with four different kinds of loss functions. For the transformer-based sentence generator, we attempt different architectures and choose the best one.

To verify whether the proposed word–sentence framework works effectively and widely, some experiments are conducted on three popular public RSIC data sets of Sydney-captions [1], UCM-captions [1], and RSICD [2]. Our word–sentence framework achieves comparable results compared with the encoder–decoder-based methods. What is more, benefiting from the visualization technique of class activation mapping (CAM) [27], each word in the sentence can be independently visualized in the image by means of attention map.

In general, our contributions to this article can be summarized as the following three points.

1) In this article, we propose a new two-stage word–sentence framework for RSIC. The proposed word–sentence framework contains two submodels of word extractor and sentence generator. Compared with the encoder–decoder framework that is a black-box model, our word–sentence framework is more interpretable and is more in accord with human understanding of image captioning.
2) To realize the word–sentence framework, for word extractor, we attempt different CNNs and loss functions to extract as many words from the remote sensing images as possible, where each word in reference sentences is seen as one category. For sentence generator, we further attempt different architectures of the transformer and choose the best one to achieve the sentence generation.
3) In order to verify the effectiveness of the proposed framework, some experiments based on different CNNs are conducted on three public RSIC data sets. Compared with some methods based on the encoder–decoder framework, our word–sentence framework still works comparably.

II. Related Work

A. Encoder–Decoder Framework

In the field of RSIC, the existing methods obey the general architecture of the encoder–decoder framework [19], which is first proposed in NIC [18], [20]–[23]. As mentioned earlier, the encoder–decoder framework consists of two submodels of encoder and decoder. As shown in Fig. 2, it is a workflow of CNN-based encoder–decoder framework.

At the encoding stage, the encoder submodel is used to detect, nd recognize the interesting features and objects from the remote sensing images, and save them in a high-dimensional semantic feature vector (or feature map). There are roughly two types of feature extraction methods for image captioning.

1) *Traditional Method-Based Encoder [2], [18]:* In this type of methods, features are handcrafted, including bag of words (BOW) [28], the Fisher vector (FV) [29], vector of locally aggregated descriptors (VLAD) [30], and scale-invariant feature transform (SIFT) [31].
2) *Deep Learning Method-Based Encoder [1], [2], [16], [19]:* In image captioning, it usually refers to convolutional neural network (CNN), including AlexNet [32], VGG [33], and ResNet [34].

The deep feature can be automatically learned by CNN with the guidance of label data. Deep feature-based methods have become mainstream due to CNN's powerful feature extraction ability. For the encoder submodel, the input is a raw RGB image, while the output is a multilayer feature map or a high-dimensional feature vector containing the semantic information in the images.

At the decoding stage, a decoder submodel plays the role of translating the feature vector of the map into a well-formed sentence. Its input is the visual feature of fixed size, and its output is the sequence output with uncertain length. To deal with the sequential problem, a decoder is usually realized by the recurrent neural network (RNN) or the long short-term memory (LSTM) [35]. As shown in Fig. 2, the words of the sentence are generated by RNN/LSTM step by step. Each word depends on the previous word and the hidden states of the RNN/LSTM cell. The generated sentence should be well-formed in grammar and logic and contain as many valuable words as possible.

The whole encoder–decoder framework is an end-to-end trainable model, which is directly optimized by the loss between the generated sentence and the reference sentence. However, the encoder–decoder framework is a lack of explainability because it works in a black-box manner. Different from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:26:20 UTC from IEEE Xplore. Restrictions apply.

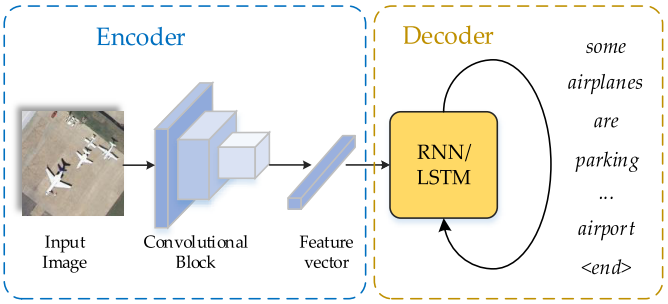


Fig. 2. Workflow of the existing CNN-based encoder–decoder framework for RSIC.

the encoder–decoder framework, the word–sentence framework of this article focuses on each valuable word and provides the visualization of its attention region.

B. Remote Sensing Image Captioning

Referring to NIC, RSIC aims to generate a sentence for a remote sensing image. However, there are some unique difficulties of remote sensing images in RSIC: a remote sensing image contains plenty of features and objects on the ground, the scale of the features and objects varies widely, and these features and objects are quite similar and mixed.

Based on the encoder–decoder framework, researchers have studied RSIC and made contributions to this field from different views. Qu *et al*. [1] release two RSIC data sets of UCM-captions and Sydney-captions and first transfer the encoder–decoder framework (CNN + RNN/LSTM) from NIC to RSIC. Shi *et al*. [16] propose a multilevel convolutional architecture for remote sensing image understanding, focusing on improving the accuracy of object recognition. Lu *et al*. [2] utilize the multimodal feature-based methods and the deep feature-based methods at the image encoding stage, publishing the largest data set of RSICD. Besides, Lu *et al*. [2] also attempt the soft and hard attention models to improve the accuracy of each word. Furthermore, Zhang *et al*. [17] introduce a multiscale image cropping and training mechanism to extract multiscale features. Recently, Zhang *et al*. [36] use an attribute attention mechanism in remote sensing images and explore the impact of the attributes in RSIC. Zhang *et al*. [37] propose visual aligning attention model for RSIC to ensure that the attention layers can accurately locate at the interesting regions Lu *et al*. [38] introduce sound activation attention mechanism to deal with the inconsistency of descriptions from different observers.

III. Word–Sentence Framework

As shown in Fig. 3, the proposed word–sentence framework is a two-stage architecture, consisting of a word extractor and sentence generator. In this section, we introduce these two submodels in detail.

A. Word Extractor

Word extractor aims to extract all the valuable words in the given remote sensing images, including noun, pronoun, numeral, adjective, conjunction, and others. In this article, as shown in Fig. 1 (left), the CNN-based multilabel classifier followed by k-max word selection strategy is used to achieve this goal.

1) CNN-Based Multilabel Classifier: For the CNN-based word extractor, its input is the raw RGB remote sensing image, and its output is the confidence vector representing the confidence degree of each word in the vocabulary. It consists of convolutional blocks, a global average pooling (GAP) layer, and a fully connected (FC) layer.

The convolutional block contains convolutional layers, a pooling layer, and a normalization layer. For each CNN models, multiple convolutional blocks are stacked to extract high-level feature map $F \in \mathbb{R}^{H_F \times W_F \times C}$ from a raw RGB image $I \in \mathbb{R}^{H_I \times W_I \times 3}$. $H_F \times W_F$ is the spatial size of feature map, and C is its channel dimensionality. $H_I \times W_I$ is the image resolution, and 3 refers to the optical RGB channels. The convolutional process is demoted as

$$F = \mathrm{CNN}(I). \tag{1}$$

There are many kinds of well-designed convolutional architectures. In this article, the convolutional blocks of four widely used CNNs of AlexNet [32], VGG16 [33], ResNet18 [34], and GoogleNet [39] are used to extract high-level semantic feature.

In order to decrease the parameters and relieve the overfitting problem, GAP [40] is used to convert the feature map F into a global feature vector $v_0 \in \mathbb{R}^C$. The kth element of v_0 is calculated by

$$v_0(k) = \frac{1}{H_F \times W_F} \sum_{i=0}^{H_F} \sum_{j=0}^{W_F} F(i, j, k). \tag{2}$$

Based on the global feature vector v_0, the multilabel confidence vector $v_1 \in \mathbb{R}^N$ can be further calculated by an FC layer, which is formulated as

$$v_1 = \mathcal{W}^T v_0 + b \tag{3}$$

where $\mathcal{W} \in \mathbb{R}^{C \times N}$ is the weighting matrix and $b \in \mathbb{R}^N$ is the bias, where N is the number of words in the vocabulary (vocabulary size). In $v_1 \in \mathbb{R}^N$, each element represents the existing confidence of the corresponding word in the vocabulary. The classifier in (3) can project the global semantic feature vector v_0 to the multilabel confidence vector v_1.

To scale the confidence of each word to range of [0, 1], the nonlinear activation function, Sigmoid, of (4) is applied in v_1. $v_2 \in \mathbb{R}^N$ is the scaled multilabel confidence vector, as shown in the following equation:

$$v_2(i) = \frac{1}{1 + e^{-v_1(i)}}. \tag{4}$$

Here, for each element in $v_2 \in \mathbb{R}^N$, its value represents the existence probability of the corresponding word in the vocabulary.

There is a word-level binary label of $y_v \in \mathbb{R}^N$ corresponding to v_2. If the ith word in the vocabulary is in the ground-truth label of the given five sentences, the value of $y_v(i)$ is equal to 1, and otherwise, it is 0.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:26:20 UTC from IEEE Xplore. Restrictions apply.

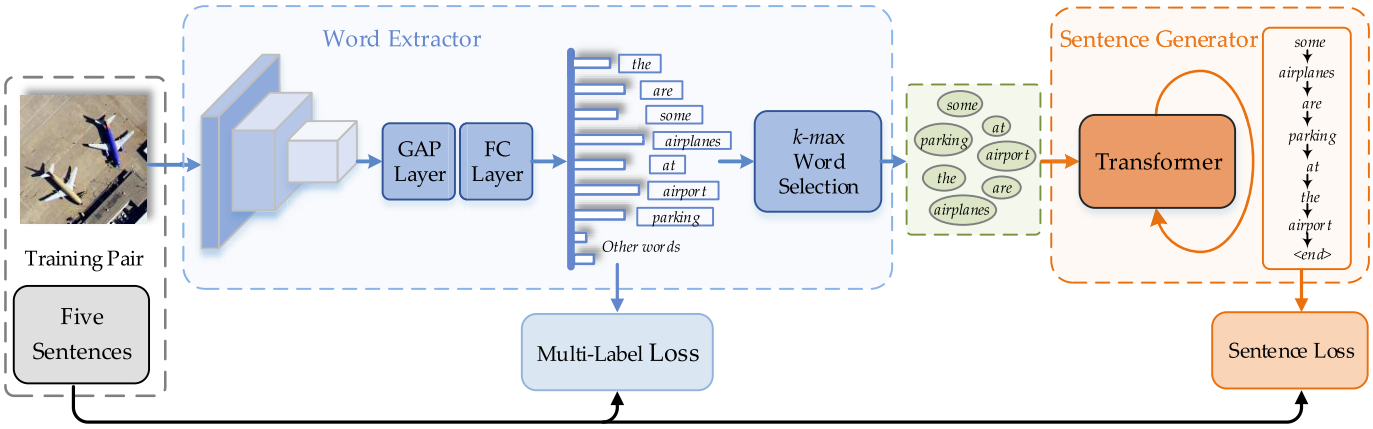


Fig. 3. Workflow of the proposed word–sentence framework for RSIC.

2) Multilabel Loss Function: It is important to design the multilabel loss function to optimize the CNN-based word extractor. In this article, we use four kinds of loss functions as multilabel loss function of mean absolute error (MAE) loss, mean square error (MSE) loss, hinge loss, and binary cross-entropy (BCE) loss [41], which are as denoted $\mathcal{L}_{\text{MAE}}$, $\mathcal{L}_{\text{MSE}}$, $\mathcal{L}_H$, and $\mathcal{L}_{\text{BCE}}$, respectively. They are formulated as

$$\mathcal{L}_{\text{MAE}} = \frac{1}{N}\sum_{i=1}^{N} ||v_2(i) - y_v(i)|| \tag{5}$$

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N} [v_2(i) - y_v(i)]^2 \tag{6}$$

$$\mathcal{L}_H = \frac{1}{N}\sum_{i=1}^{N} [1 - y_v(i) * v_2(i)] \tag{7}$$

$$\mathcal{L}_{\text{BCE}} = \frac{1}{N}\sum_{i=1}^{N} \{y_v(i) * \log[v_2(i) + \delta] + [1 - y_v(i)] * \log[1 - v_2(i) + \delta]\}. \tag{8}$$

$\mathcal{L}_{\text{MAE}}$ and $\mathcal{L}_{\text{MSE}}$ are usually used for regression task; however, they can also be used to deal with the classification problem. $\mathcal{L}_H$ and $\mathcal{L}_{\text{BCE}}$ are specially designed for binary classification task. In $\mathcal{L}_{\text{BCE}}$, δ is a very small positive number of 1e-5, which is used to avoid log0.

3) k-Max Word Selection: To decrease the computational complexity, the first k maximum-probability words are selected from the vocabulary confidence of v_2 as a BOW $\mathbf{w} = \{w_1, \ldots, w_k\}$, where $w_i \in \mathbb{R}^E$ is the embedding of the ith word. This translation procedure can be summarized as Algorithm 1.

In Algorithm 1, v_2 is sorted with descending order in line 2. Then, the first k maximum-probability words in v_2 are selected as the reference for sentence generation from lines 3–5. In line 4, because the confidence value of each word is in the range of [0, 1], 0.5 is used as the threshold to binary the confidence into either 0 or 1. In line 5, word bedding operation is used to embed the extracted words into word vector space.

Algorithm 1 k-Max Word Selection

Input: The vector of word confidence, v_2;
The number of the one-hot words, k;
Output: A bag of k-max words, $\mathbf{w}$;
1: All the elements of w are initialized by 0;
2: $value, index = descending_sorting(v_2)$;
3: **for** $i = 0 \rightarrow k$ **do**
4: **if** $value[i] > 0.5$ **then**
5: $w_i = embedding(index[i])$
6: **return w**

B. Sentence Generator

Although the most valuable words $\mathbf{w} = (w_1, \ldots, w_k)$ in images are extracted by word extractor, they only are sorted according to the probability of existence. The grammatical and logical relationship between them has not been established. To achieve this goal, the transformer [42], which can handle the sequence-to-sequence problem, is used as the sentence generator to translate the disordered words into a well-formed sentence.

1) Transformer-Based Sentence Generation: In the word–sentence framework, the transformer illustrated in Fig. 4 is composed of encoder and decoder. First, transformer's encoder maps the word embeddings $\mathbf{w} = \{w_1, \ldots, w_k\}$ into a sequence of memory states $\mathbf{z} = \{z_1, \ldots, z_k\}$, where $z_i \in \mathbb{R}^E$. Second, based on $\mathbf{z}$, transformers' decoder can generate an output sequence $\mathbf{y} = \{y_1, \ldots, y_k\}$ step by step, where $y_i \in \mathbb{R}^E$. At time step t, the decoder generates the current output $\{y_1, \ldots, y_t\}$ according to the combination of the previous output $\{y_1, \ldots, y_{t-1}\}$ and memory states $\mathbf{z}$. In order to keep the length of different time steps consistent during the training stage, the rest sequence of $\{y_t, \ldots, y_k\}$ is padded by zero for parallel training. Finally, the output sequence $\mathbf{y}$ is classified into the word probability sequence $\mathbf{s} = \{s_1, \ldots, s_k\}$, where $s_i \in \mathbb{R}^N$.

The encoder of transformers is a stack of several identical encoder layers. For each encoder layer, there are two sublayers of multihead attention and an FC feed-forward network. For these two sublayers, the operations of residual connection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:26:20 UTC from IEEE Xplore. Restrictions apply.

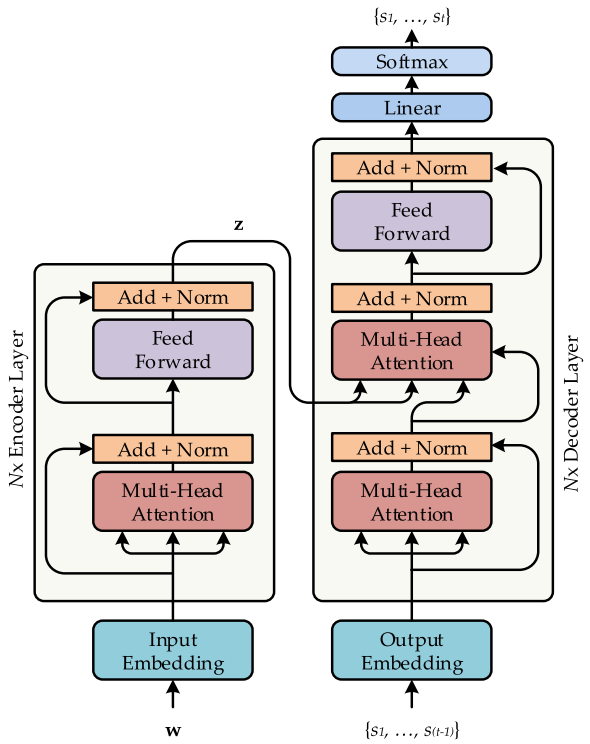


Fig. 4. Illustration of the transformer as the sentence generator.

and batch normalization [43] are inserted. The decoder of transformers is also a stack of several identical decoder layers. For each decoder layer, there are three sublayers. Apart from the multihead attention and an FC feed-forward network in the encoder layer, another multihead attention is inserted to absorb the memory states $\mathbf{z}$. The same residual connection as the encoder layer is also used in the decoder layer.

For multihead attention, given input $\mathbf{x}_{\text{att}}$, it generates output $\mathbf{y}_{\text{att}}$ by

$$\mathbf{x}_{\text{res}} = \text{MultiHeadAtt}(\mathbf{x}_{\text{att}}) \tag{9}$$

$$\mathbf{y}_{\text{att}} = \text{Norm}(\mathbf{x}_{\text{att}} + \mathbf{x}_{\text{res}}) \tag{10}$$

where MultiHeadAtt is the concatenation of multiple self-attention heads. In order to prevent leakage of sequential data and keep the length of different time steps consistent, in $\mathbf{x}$, the words appearing after the current time step are padded by zero.

For the feed-forward network, given input $\mathbf{x}_{\text{feed}}$, it generates output $\mathbf{y}_{\text{feed}}$ by

$$\mathbf{x}_{\text{res}} = f_2(\text{Relu}(f_1(\mathbf{x}_{\text{feed}}))) \tag{11}$$

$$\mathbf{y}_{\text{feed}} = \text{Norm}(\mathbf{x}_{\text{feed}} + \mathbf{x}_{\text{res}}) \tag{12}$$

where Relu is a nonlinear activation function. f_1 and f_2 are two FC layers, where f_1 maps the input into a high-dimensional space and f_2 restores it to the original dimension.

At the end of the transformer, the output sequence $\mathbf{y}$ is mapped to a sentence $\mathbf{s} = \{s_1, \ldots, s_k\}$, where $s_t \in \mathbb{R}^N$ is the generated word at time step t. The mapping procedure is given as

$$\bar{s}_i = f_s(y_i) \tag{13}$$

$$s_i(j) = \frac{e^{\bar{s}_i(j)}}{\sum_{i=1}^{N} e^{\bar{s}_i(j)}} \tag{14}$$

where f_s is a single FC layer, and (14) is softmax operation to make the probability of all the words summed to 1.

2) Sentence Loss: For the generated sentence $\mathbf{s}$, there is the corresponding label $\mathbf{y} = \{y_1, \ldots, y_k\}$, where y_t is index of the ground-truth word at time t. At the training stage, the transformer is optimized to minimize the following loss function between $\mathbf{s}$ and $\mathbf{y}$:

$$\mathcal{L}(s_t, y_t) = -\log \frac{e^{s_t[y_t]}}{\sum_{i=1}^{N} e^{s_t[j]}}. \tag{15}$$

IV. Experiments

In this section, the image captioning data sets and evaluation metrics used in this article are introduced first. Then, the experimental details are provided. Following that, some ablation experiments of word extractor and sentence generator are conducted on three data sets, and some samples of the visualization of words in the generated sentence are provided. Finally, we report experimental results in comparison with some other encoder–decoder-based methods.

A. Data Sets and Evaluation Metrics

In this article, experiments are performed on three public RSIC data sets.

Sydney-Captions [1]: It is based on the remote sensing scene data set of the Sydney Data set [46]. The Sydney data set contains 613 images of seven classes, including residential, airport, meadow, rivers, ocean, industrial, and runway. Each image measures 500 × 500 pixels with a pixel resolution of about 50 cm. All these images were manually extracted from the image of Sydney, which is download on Google Earth with a size of 18 000 × 14 000 pixels.

UCM-Captions [1]: It is based on the remote sensing scene classification data set of the UC Merced (UCM) Land Used data [47], which has 2100 images of 21 typical land-use scene classes, including agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium-density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis courts. Each class has 100 images measuring 256 × 256 pixels with a pixel resolution of 30 cm in the RGB space. The UCM data set is extracted from aerial orthography and downloaded from the United States Geological Survey.

RSICD [2]: It contains a total of 10 921 remote sensing images in different kinds of areas. All the images in this data set come from Google Earth and measure 224 × 224 pixels with different resolutions. So far, it is the largest data set published in the field of RSIC. There are 30 kinds of scenes: airport, bridge, beach, baseball field, open land, commercial, center, church, desert, dense residential, forest, farmland, industrial, mountain, medium residential, meadow, port, pond,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:26:20 UTC from IEEE Xplore. Restrictions apply.

TABLE I

RESULTS OF BLEU1 ON THREE RSIC DATA SETS USING DIFFERENT CNN BACKBONES AND LOSS FUNCTIONS. B1: BLEU1. $\mathcal{L}_{\text{MAE}}$: MAE LOSS. $\mathcal{L}_{\text{MSE}}$: MSE LOSS. $\mathcal{L}_H$: HINGE LOSS. $\mathcal{L}_{\text{BCE}}$: BCE LOSS

Data Set	CNN Backbones	B1 on $\mathcal{L}_{MAE}$	B1 on $\mathcal{L}_{MSE}$	B1 on $\mathcal{L}_H$	B1 on $\mathcal{L}_{BCE}$
Sydney-captions	AlexNet	56.14	**72.73**	69.52	72.42
	VGG16	70.26	**73.47**	69.37	72.51
	ResNet18	75.50	74.67	**76.23**	75.73
	GoogleNet	**75.46**	74.51	75.45	75.24
UCM-captions	AlexNet	64.57	67.80	69.02	**71.25**
	VGG16	40.33	**70.85**	36.32	69.32
	ResNet18	77.40	**78.18**	77.63	77.42
	GoogleNet	78.19	78.02	**78.27**	77.99
RSICD	AlexNet	67.31	**78.01**	67.18	76.38
	VGG16	69.12	**76.97**	38.89	76.27
	ResNet18	62.03	**78.03**	63.85	76.81
	GoogleNet	62.41	**77.96**	63.69	77.67

TABLE II

SPECIFIC SETTINGS OF THE TRANSFORMERS OF DIFFERENT SIZES. TE: TRANSFORMER-ENCODER. TD: TRANSFORMER-DECODER

Transformer Size	s1	s2	s3	s4
Number of TE&TD layers	1	2	4	8
Dimension of input&output	64	128	256	512
Dimension of feedforward	256	512	1024	2048

parking, park, playground, river, railway station, resort, storage tanks, stadium, sparse residential, square, school, and viaduct.

For each image in these three data sets, five sentences from different observers are provided. The split of training, validation, and test of these data sets follows the original literature. They are all split by the same ratio of 80%/10%/10% on the training/validation/test set.

It is an important issue to evaluate the quality of the generated caption for a given image, and there are many evaluation metrics applied in image captioning. In this article, the evaluation metrics for RSIC are as follows.

BLEU: Bilingual Evaluation Understudy (BLEU) [48] is first proposed in the field of machine translation and used to measure the matching degree of n-grams (n continuous words) between the generated text and the reference text. In this article, n is set to 1, 2, 3, and 4, which corresponds to BLEU1, BLEU2, BLEU3, and BLEU4, respectively. BLEU pays attention to the accuracy of the n-grams words in the generated sentence. It is a simple, fast, and effective evaluation metric with a good performance.

CIDEr: Different from BLEU that comes from the task of machine translation belonging to NLP, the Consensus-based Image Description Evaluation (CIDEr) [49] is especially proposed for the task of image captioning. It is an automatic consensus metric. The characteristic of CIDEr is to weight n-grams in accordance with their frequency in the whole data set and decrease the weight of the noncritical words in the generated sentence.

ROUGE-L: Recall-Oriented Understudy for Gisting Evaluation (ROUGE) [50] is a set of metrics concentrating on the recall of captions and is used for text summary. Different types of ROUGE, such as ROUGE-1, ROUGE-2, ROUGE-L, and ROUGE-W, are used for different tasks. ROUGE-L is a metric that calculates the F-measure given the longest common subsequence (LCS) and is used to evaluate the quality of RSIC in experiments.

METEOR: Metric for Evaluation of Translation with Explicit ORdering (METEOR) [51] is measured by calculating an alignment between the generated and reference sentence. Based on a single-precision weighted harmonic mean and single-word recall rate, METEOR takes into consideration both the precision and recall rate; therefore, it can handle some of the defects inherent in the BLEU standard.

In conclusion, there are seven kinds of metrics of BLEU1, BLEU2, BLEU3, BLEU4, CIDEr, ROUGE-L, and METEOR used to comprehensively evaluate the generated captions of remote sensing images in this article. The higher scores the generated captions get, the better quality they have.

B. Experimental Details

Framework Settings: In the word–sentence framework, the CNN-based word extractor and transformer-based sentence generator are separately optimized. For the CNN-based word extractor, we attempt four widely used CNN architectures pretrained in ImageNet [52], including AlexNet [32], VGG16 [33], ResNet18 [34], and GoogleNet [39]. All the classifiers of the CNNs are replaced by the single FC layer followed by the Sigmoid nonlinear activation function. For the transformer-based sentence generator, the head number of multihead attention is set to 8, the number of transformer encoder layer and transformer decoder layer is set from 1 to 8, the dimension of the input and output is set from 64 to 512, and the forward dimension is set from 256 to 2048.

Training Settings: In experiments, all the training and test images are resized to 224 $\times$ 224, and the training images are randomly horizontally flipped with the 50% probability for data augmentation. The number of the extracted words and the max length of the generated sentence, k, is set to 30. Adam is used as the optimizer for both word extractor and sentence generator with the learning rate set to 1e-4. All

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:26:20 UTC from IEEE Xplore. Restrictions apply.

TABLE III
RESULTS ON THREE DATA SETS BASED ON DIFFERENT SIZES OF TRANSFORMER

Data Set	Word-Sentence Framework	BLEU1	BLEU2	BLEU3	BLEU4	CIDEr	ROUGE_L	METEOR
Sydney-captions	ResNet18 + Transformer_s1	78.25	70.09	62.09	54.96	1.7756	0.6814	0.3914
	ResNet18 + Transformer_s2	**78.91**	**70.94**	63.17	56.25	2.0411	**0.6922**	**0.4181**
	ResNet18 + Transformer_s3	78.23	70.12	**64.17**	**59.45**	**2.2136**	0.6741	0.4129
	ResNet18 + Transformer_s4	74.13	66.02	59.09	53.11	1.6968	0.6396	0.3818
UCM-captions	GoogleNet + Transformer_s1	77.39	69.57	64.07	59.45	2.6698	0.6967	0.4201
	GoogleNet + Transformer_s2	**79.31**	**72.37**	**66.71**	**62.02**	**2.7871**	**0.7132**	**0.4395**
	GoogleNet + Transformer_s3	78.08	70.36	64.57	59.56	2.6446	0.7068	0.4293
	GoogleNet + Transformer_s4	77.10	69.30	63.36	58.67	2.5703	0.7073	0.4219
RSICD	ResNet18 + Transformer_s1	71.13	57.82	48.65	41.84	2.0020	0.6233	0.3069
	ResNet18 + Transformer_s2	70.77	56.84	47.79	41.20	1.9547	0.6069	0.3076
	ResNet18 + Transformer_s3	**72.40**	**58.61**	**49.33**	**42.50**	**2.0629**	**0.6260**	**0.3197**
	ResNet18 + Transformer_s4	71.96	58.07	48.85	42.11	2.0243	0.6192	0.3165

TABLE IV
COMPARISON OF SOME STATE-OF-THE-ART METHODS ON SYDNEY-CAPTIONS

Models	BLEU1	BLEU2	BLEU3	BLEU4	CIDEr	ROUGE_L	METEOR
VLAD-LSTM [2]	49.13	34.72	27.60	23.14	0.9164	0.4201	0.1930
VGG16-RNN [1]	51.30	37.50	20.40	19.30	0.3220	—	0.1850
VGG16-LSTM [1]	54.60	39.50	22.30	21.20	0.3720	—	0.2050
CSMLF(ft) [44]	59.98	45.83	38.69	34.33	0.9378	0.5018	0.2475
SIFT-LSTM [2]	57.93	47.74	41.83	37.40	0.9873	0.5366	0.2707
Sound-f-a [38]	71.55	63.23	54.69	46.60	1.8027	0.6035	0.3132
VAA [37]	74.31	66.46	60.29	54.95	**2.4073**	**0.6999**	0.3930
Soft Attention-Based GoogleNet [2]	71.28	62.39	55.27	49.24	2.0343	0.6913	0.3675
Hard Attention-Based GoogleNet [2]	76.89	66.13	58.40	51.70	1.9863	0.6842	0.3719
Our Word_Sentence Framework	**78.91**	**70.94**	**63.17**	**56.25**	2.0411	0.6922	**0.4181**

TABLE V
COMPARISON OF SOME STATE-OF-THE-ART METHODS ON UCM-CAPTIONS

Models	BLEU1	BLEU2	BLEU3	BLEU4	CIDEr	ROUGE_L	METEOR
PCSMLF [45]	43.61	27.28	18.55	12.10	0.2227	0.3927	0.1320
SIFT-LSTM [2]	55.17	41.66	34.89	30.40	1.3603	0.5235	0.2432
VLAD-LSTM [2]	70.16	60.85	54.96	50.30	2.3131	0.6520	0.3464
VGG16-RNN [1]	60.10	50.70	32.80	20.80	0.4280	—	0.1930
VGG16-LSTM [1]	63.80	53.60	37.70	21.90	0.4510	—	0.2060
Sound-a-f [38]	78.28	72.76	67.59	63.33	3.2906	0.6864	0.3803
RTRMN(statistical) [44]	80.28	73.22	68.21	63.93	3.1270	0.7726	0.4258
VAA [37]	81.92	75.11	69.27	63.87	**3.3946**	**0.7824**	0.4380
Soft Attention-Based GoogleNet [2]	76.36	67.66	61.03	55.37	2.8567	0.7400	0.4010
Hard Attention-Based GoogleNet [2]	**83.75**	**76.22**	**70.42**	**65.62**	3.2001	0.7962	**0.4489**
Our Word_Sentence Framework	79.31	72.37	66.71	62.02	2.7871	0.7132	0.4395

the models are trained for 50 epochs, and the size of the minibatch is 64. In addition, all the experiments in this article are implemented by Pytorch 1.3.0 in the computing equipment of 64-GB memory CPU and 1 × 12 GB memory GPU of NVIDIA GeForce GTX 1080Ti.

C. Experiments on Word Extractor

In this section, some ablation experiments of word extractor are performed to explore the influence of CNN backbones and the multilabel loss function on word extraction. The BLEU1 is used to evaluate the quality of the extracted words.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:26:20 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISON OF SOME STATE-OF-THE-ART METHODS ON RSICD

Models	BLEU1	BLEU2	BLEU3	BLEU4	CIDEr	ROUGE_L	METEOR
SIFT-LSTM [2]	48.59	30.33	21.86	16.78	1.0528	0.4174	0.1966
VLAD-LSTM [2]	50.04	31.95	23.19	17.78	1.1801	0.4334	0.2046
CCSMLF [45]	57.59	39.59	28.32	22.17	0.5297	0.4455	0.2128
VGG19-LSTM [2]	58.33	42.26	33.10	27.02	2.0332	0.5189	0.2613
RTRMN(statistical) [44]	61.02	45.14	35.35	28.59	1.4820	0.5452	0.2751
text-a-a [38]	65.05	51.32	41.44	33.61	1.6828	0.5268	0.2909
Soft Attention-Based GoogleNet [2]	67.37	53.03	43.24	35.98	1.9652	0.6212	**0.3339**
Hard Attention-Based GoogleNet [2]	68.81	54.52	44.70	37.25	2.0215	**0.6284**	0.3322
Our Word_Sentence Framework	**72.40**	**58.61**	**49.33**	**42.50**	**2.0629**	0.6260	0.3197

For CNN backbones, AlexNet, VGG16, ResNet18, and GoogleNet are the quite representative CNN models so far. AlexNet is the most classical convolutional network that attracts researchers' attention to deep learning. VGG16 is the first very deep convolutional network and obtains excellent results in image recognition. ResNet18 accelerates the convergence of network parameters and improves the performance due to its residual connection. GoogleNet applies the multiscale convolutional kernels and, therefore, has the ablation of multiscale feature representation, which is quite helpful for remote sensing images. For loss function, $\mathcal{L}_{\text{MAE}}$ and $\mathcal{L}_{\text{MSE}}$ are two of the most widely used loss function used for not only regression tasks but also classification tasks. $\mathcal{L}_H$ and $\mathcal{L}_{\text{BCE}}$ are specially designed for binary classification task.

The cross results of CNN backbones and multilabel loss functions are shown in Table I. In all the CNN backbones, GoogleNet and ResNet have the best and the most stable performance. Limited by the shallow network structure, AlexNet cannot extract the complex information, including not only object words but also nonobject words, describing the relationship of these objects. For VGG16, it has unsatisfactory performance on the small data sets of Sydney-caption and UCM-captions and is quite sensitive to the loss function on all the data sets. In the four multilabel loss functions, overall, $\mathcal{L}_H$ performs badly across and CNNs and is not stable on different data sets. It is obvious that it is not suitable for word extraction in RSIC. The performance of $\mathcal{L}_{\text{MAE}}$ decreases rapidly with an increase in the size of data sets, which is mainly caused by the unstable gradient in $\mathcal{L}_{\text{MAE}}$. $\mathcal{L}_{\text{BCE}}$ and $\mathcal{L}_{\text{MSE}}$ have similar performance overall and are more effective when the data set is large. Overall, the performance of VGG16 and AlexNet is not stable when they are optimized by $\mathcal{L}_{\text{MSE}}$ and $\mathcal{L}_H$. As far as we are concerned, it is probably caused by the discontinuous gradient of $\mathcal{L}_{\text{MSE}}$ and $\mathcal{L}_H$, and VGG16 and AlexNet do not have the skip connection or multiscale convolution of ResNet18 and GoogleNet that are beneficial for the stable training.

In general, both CNN backbones and loss functions have an important influence on the word extraction. According to the experimental results, in the following experiments, ResNet optimized by the best loss function is used as the word extractor on Sydney-captions and RSICD, while GoogleNet optimized by the best loss function is used as the word extractor on UCM-captions.

D. Experiments on Sentence Generator

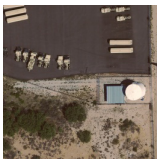
In this section, some ablation experiments about the transformer-based sentence generator are conducted to study the influence of the transformer size. The experiments are conducted based on four kinds of transformer sizes, which are summarized in Table II.

Table III reports the experimental results of the above four kinds of transformers on three data sets. When the size of the transformer is too small, the sentence generator is underfitting, and therefore, there is not enough network capacity to represent the complex pattern of sentence generation. When the size of the transformer is too large, the sentence generator falls into overfitting, and some noise would be learned, which is also bad for the quality of the generated sentence. According to the results in the table, overall, Transformer_s2 achieves the best performance in the word–sentence framework for RSIC, obtaining the most of the optimal scores of all kinds of metrics on all the data sets. Besides the performance, Transformer_s2 has another advantage in the scale of network parameters, which is also an important reference for model evaluation. When comparing the BLEU1 score between Tables I and III, it could be found that there are decreases of different degrees of BLEU1 on these three data sets when the disordered words are organized by sentence generator. It indicates that some useful words are lost or some useless words are produced during the sentence generation.

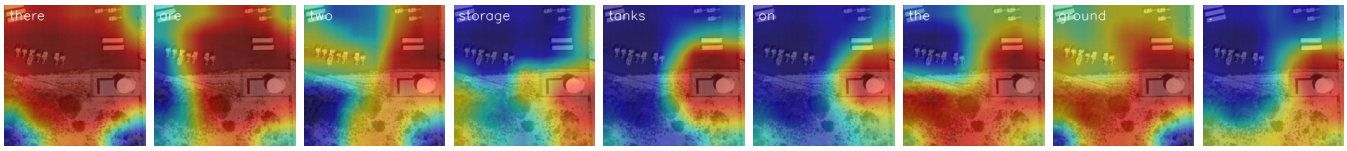
It is worth mentioning that, in the following experiments, the word–sentence framework refers to the combination of ResNet18 and Transformer_s2 Sydney-captions, the combination of GoogleNet and Transformer_s2 for UCM-captions, and the combination of GoogleNet and Transformer_s3 for RSICD.

In order to intuitively show the workflow of the proposed word–sentence framework, some samples of words, sentences, and attention maps are shown in Fig. 5. In the figure, there are three samples of the disordered words and the well-formed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:26:20 UTC from IEEE Xplore. Restrictions apply.

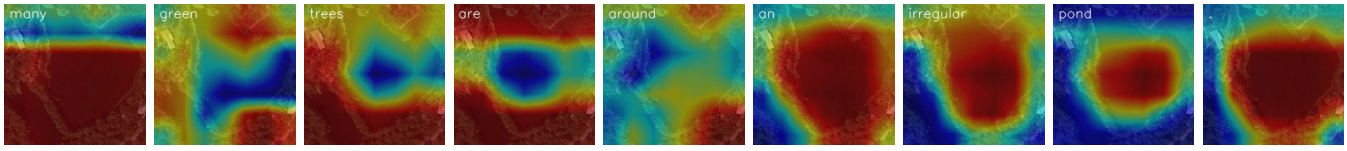


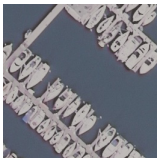
Words: some, are, storage, there, a, two, with, and, is, the.
Sentence: there are two storage tanks on the ground.
Reference: (1): A small storage tank is on the ground. (2): there is one storage tank on the ground and some buildings beside. (3)-(4): a storage tank is on the ground and some buildings beside. (5): There is a white storage tank on the ground.





Words: pond, green, many, around, a, are, trees.
Sentence: many green trees are around an irregular pond.
Reference: (1): it is a pond with dark green water in the middle. (2): the pond is surrounded by green plants. (3): many green trees are around two irregular ponds. (4): many green trees are around two irregular ponds. (5): it is a pond with dark green water in the middle.





Words: positions, harbor, are, blue, water, many, docked, of, and, is, deep, at, boats, neatly, lots, the.
Sentence: lots of boats docked at the harbor and the water is deep blue.
Reference: (1): lots of boats docked at the harbor and the water is deep blue. (2): lots of boats docked neatly at the harbor. (3): many boats docked neatly at the harbor and the water is deep blue. (4): many boats docked neatly at the harbor. (5): lots of boats docked neatly at the harbor and the boats are colsed to each other.

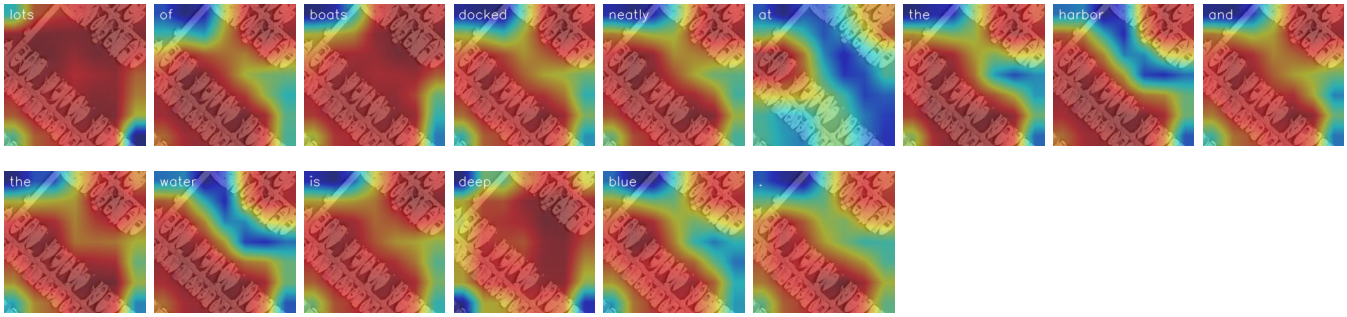


Fig. 5. Visualization of each word in the generated sentence using CAM.

sentences, which are, respectively, provided by the word extractor and the sentence generator. In Sample I, there is a wrong word of "two" extracted by the word extractor, which further leads to the imprecise description of "two storage tanks" in the sentence. In Sample II, there is an extra word of "irregular" produced by sentence generator, which is not in the words. In Sample III, a useless word of "positions" is produced by the word extractor, but it is dropped by the sentence generator. These three samples indicate that the sentence generator has multiple functions of sorting the disordered words, dropping the useless words, and generating the auxiliary words.

Besides the words and sentences, the visualization in images of each word in the generated sentences is also provided in the figure using the classwise attention technique of CAM [27]. Although most of the valuable words are extracted from the image, their attention maps are not always in accordance with the correct areas. It may be caused by the following three aspects: 1) there are not enough data for word-level multiclass classification, especially for the words of low frequency; 2) some words of conjunction, preposition, adjective, and numeral have no clear physical meaning and, therefore, are hard to locate; and 3) some words are usually symbiotic, such as "harbor," "boat," and "water." Without the accurate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:26:20 UTC from IEEE Xplore. Restrictions apply.



Fig. 6. Examples of test images and the corresponding generated captions. (a) There are two straight freeways with some plants beside them. (b) Small baseball diamond. (c) Waves slapping a white sand beach. (d) There are some buildings with cars parked beside them. (e) Residential area with houses arranged neatly and divided into rectangles by some roads. (f) River with some green trees in two sides of it. (with lack of ***bridge***). (g) There are ***two*** airplanes at the airport. (h) ***Road*** goes through this area. (i) Many buildings and green trees are in a ***school***. (j) Curved river with deep green water (with lack of *plants*). (k) Several buildings are near a ***church***. (l) There is a small ***storage tank on the ground***. (m) There is a small ***tennis court*** surrounded by some plants. (n) There are lots of ***buildings***. (o) There are some white marking lines on the runways while a ***lawn*** is beside.

bounding box label, it is hard for CNNs to distinguish them and locate their accurate areas.

More representative samples of RSIC provided by the proposed word–sentence framework are shown in Fig. 5. From Fig. 5(a)–(d), they are correct and well-formed in logic and grammar. From Fig. 5(e)–(h), there are some not fatal errors in the generate sentences, such as an imprecise numeral, lacking objects, and ambiguous descriptions. From Fig. 5(i)–(l), there are some fatal errors that the main objects in the images are recognized by mistake.

E. Comparison With State-of-the-Art Methods

In order to evaluate the proposed word–sentence framework objectively, it is necessary to make a comparison with some state-of-the-art results of encoder–decoder-based methods.

1) Sydney-Captions: First, we compare the proposed word–sentence framework with some state-of-the-art methods of the encoder–decoder framework on the smallest RSIC data set of Sydney-captions, and the results are reported in Table IV. For SIFT-LSTM [2] and VLAD-LSTM [2], their encoders belong to traditional feature extraction methods, and their performance is worst. VGG16-RNN [1] and VGG16-LSTM [1] share the CNN-based encoder of VGG16 but are different in the decoder. Their performance is better than SIFT-LSTM and VLAD-LSTM. Soft/hard attention-based GoogleNet [2] introduces the attention mechanism and obtains an obvious improvement. Sound-f-a [38] and VAA [37] have a similar and excellent performance. It could be found that our method has an obvious advantage in the scores of BLEU1-3 and CIDEr and just falls behind others in ROUGE_L. Overall, our word–sentence framework outperforms the existing encoder–decoder-based methods, especially the VGG16-RNN/LSTM that is the encoder–decoder framework without any improvement strategy. Such results demonstrate that the proposed word–sentence works well in the field of RSIC when the data set is on a smaller scale.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:26:20 UTC from IEEE Xplore. Restrictions apply.

2) UCM-Captions: Second, we make a comparison between the proposed framework and other methods, and the results are reported in Table V. The methods used in UCM-captions are the same as those in Sydney-captions. According to the results, our framework is still far better than the typical encoder–decoder-based methods of SIFT/VLAD-LSTM and VGG16-RNN/LSTM and just slightly falls behind the hard attention-based GoogleNet. Although our framework does not get the best performance, it is still comparable with the encoder–decoder-based methods.

3) RSICD: Finally, the comparison is made on the largest data set of RSICD, and the results are provided in Table VI. In RSICD, our framework obtains the best score of BLEU1, CIDEr, and ROUGE_L in the table. However, for BLEU2–BLEU4 that evaluate the matching degree of n continuous words (n-grams and $n \geqslant 2$), our method is not the hard attention-based GoogleNet [2]. In hard attention-based GoogleNet, location information is retained and sent into the decoder submodel. In our framework, however, the location information of the extracted words is lost, and their relationship is weakened. Therefore, its n-grams is not hard attention-based GoogleNet. It is a problem to be solved in the proposed encoder–decoder framework.

In general, the proposed word–sentence framework is comparable with the state-of-the-art methods; all of them obey the encoder–decoder framework. Compared with the encoder–decoder framework, the word–sentence framework is more in line with human understanding of the logic from an image to words and then from words to a well-formed sentence. It is a potential framework because there are some clear points to improve, such as utilizing the attention mechanism to improve the quality of the extracted words.

V. Conclusion

In this article, a novel explainable word–sentence framework is proposed for RSIC. The proposed word–sentence framework consists of the word extractor, which extracts the useful words from images as many as possible, and the sentence generator, which organizes these words into a well-formed sentence. For word extractor, it is realized by the CNN-based multilabel classifier, and some experiments are conducted to study the influence of CNN architectures and multilabel loss function. For the sentence generator, it is achieved by the transformer that can deal with the sequence-to-sequence problem, and some ablation experiments are also conducted to explore the impact of the transformer. Finally, our word–sentence framework achieves comparable results in comparison with some existing methods; all of them follow the encoder–decoder framework. In future work, we will focus on improving the quality of the words produced by the word extractor, which is the basis of the sentence generator.

References

[1] B. Qu, X. Li, D. Tao, and X. Lu, "Deep semantic understanding of high resolution remote sensing image," in *Proc. Int. Conf. Comput., Inf. Telecommun. Syst. (CITS)*, Jul. 2016, pp. 1–5.

[2] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018.

[3] W. Huang, Q. Wang, and X. Li, "Feature sparsity in convolutional neural networks for scene classification of remote sensing image," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 3017–3020.

[4] Q. Wang, S. Liu, J. Chanussot, and X. Li, "Scene classification with recurrent attention of vhr remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1155–1167, Feb. 2019.

[5] T. R. Martha, N. Kerle, C. J. van Westen, V. Jetten, and K. V. Kumar, "Segment optimization and data-driven thresholding for knowledge-based landslide detection by object-based image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 4928–4943, Dec. 2011.

[6] G. Cheng, L. Guo, T. Zhao, J. Han, H. Li, and J. Fang, "Automatic landslide detection from remote-sensing imagery using a scene classification method based on BoVW and pLSA," *Int. J. Remote Sens.*, vol. 34, no. 1, pp. 45–59, Jan. 2013.

[7] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[8] S. Bhagavathy and B. S. Manjunath, "Modeling and detection of geospatial objects using texture motifs," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 12, pp. 3706–3715, Nov. 2006.

[9] X. Lu, X. Zheng, and X. Li, "Latent semantic minimal hashing for image retrieval," *IEEE Trans. Image Process.*, vol. 26, no. 1, pp. 355–368, Jan. 2017.

[10] Y. Yang and S. Newsam, "Geographic image retrieval using local invariant features," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 2, pp. 818–832, Feb. 2013.

[11] X. Li, X. Zhang, W. Huang, and Q. Wang, "Truncation cross entropy loss for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, early access, Jul. 30, 2020, doi: 10.1109/TGRS.2020.3010106.

[12] W. Huang, Q. Wang, and X. Li, "Denoising-based multiscale feature fusion for remote sensing image captioning," *IEEE Geosci. Remote Sens. Lett.*, early access, Apr. 6, 2020, doi: 10.1109/LGRS.2020.2980933.

[13] G. Meinel and M. Neubert, "A comparison of segmentation programs for high resolution remote sensing data," *Int. Arch. Photogramm. Remote Sens.*, vol. 35, pp. 1097–1105, 2004.

[14] J. Yuan, D. Wang, and R. Li, "Remote sensing image segmentation by combining spectral and texture features," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 16–24, Jan. 2014.

[15] X. Zhang, W. Huang, Q. Wang, and X. Li, "SSR-NET: Spatial-spectral reconstruction network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, early access, Sep. 3, 2020, doi: 10.1109/TGRS.2020.3018732.

[16] Z. Shi and Z. Zou, "Can a machine generate humanlike language descriptions for a remote sensing image?" *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3623–3634, Jun. 2017.

[17] X. Zhang, Q. Wang, S. Chen, and X. Li, "Multi-scale cropping mechanism for remote sensing image captioning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 10039–10042.

[18] V. Ordonez, G. Kulkarni, and T. L. Berg, "Im2text: Describing images using 1 million captioned photographs," in *Proc. Adv. Neural Inf. Process. Syst.*, 2011, pp. 1143–1151.

[19] O. Vinyals, A. Toshev, S. Bengio, and D. Erhan, "Show and tell: A neural image caption generator," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3156–3164.

[20] K. Xu *et al.*, "Show, attend and tell: Neural image caption generation with visual attention," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 2048–2057.

[21] Q. You, H. Jin, Z. Wang, C. Fang, and J. Luo, "Image captioning with semantic attention," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4651–4659.

[22] J. Mao, W. Xu, Y. Yang, J. Wang, Z. Huang, and A. Yuille, "Deep captioning with multimodal recurrent neural networks (m-RNN)," 2014, *arXiv:1412.6632*. [Online]. Available: http://arxiv.org/abs/1412.6632

[23] W. Jiang, L. Ma, X. Chen, H. Zhang, and W. Liu, "Learning to guide decoding for image captioning," in *Proc. 32nd AAAI Conf. Artif. Intell.*, 2018, pp. 1–8.

[24] J. Song, Y. Guo, L. Gao, X. Li, A. Hanjalic, and H. T. Shen, "From deterministic to generative: Multimodal stochastic RNNs for video captioning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 10, pp. 3047–3058, Oct. 2019.

[25] L. Gao, X. Li, J. Song, and H. T. Shen, "Hierarchical LSTMs with adaptive attention for visual captioning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 5, pp. 1112–1131, May 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:26:20 UTC from IEEE Xplore. Restrictions apply.

[26] J. Song, T. He, L. Gao, X. Xu, A. Hanjalic, and H. T. Shen, "Unified binary generative adversarial network for image retrieval and compression," *Int. J. Comput. Vis.*, early access, pp. 1–22, Feb. 2020.
[27] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.
[28] J. Sivic and A. Zisserman, "Video google: A text retrieval approach to object matching in videos," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2003, p. 1470.
[29] F. Perronnin, Y. Liu, J. Sanchez, and H. Poirier, "Large-scale image retrieval with compressed Fisher vectors," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 3384–3391.
[30] H. Jegou, F. Perronnin, M. Douze, J. Sanchez, P. Perez, and C. Schmid, "Aggregating local image descriptors into compact codes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 9, pp. 1704–1716, Sep. 2012.
[31] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.
[32] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.
[33] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*. [Online]. Available: http://arxiv.org/abs/1409.1556
[34] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[35] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
[36] X. Zhang, X. Wang, X. Tang, H. Zhou, and C. Li, "Description generation for remote sensing images using attribute attention mechanism," *Remote Sens.*, vol. 11, no. 6, p. 612, Mar. 2019.
[37] Z. Zhang, W. Zhang, W. Diao, M. Yan, X. Gao, and X. Sun, "VAA: Visual aligning attention model for remote sensing image captioning," *IEEE Access*, vol. 7, pp. 137355–137364, 2019.
[38] X. Lu, B. Wang, and X. Zheng, "Sound active attention framework for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1985–2000, Mar. 2020.
[39] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.
[40] M. Lin, Q. Chen, and S. Yan, "Network in network," 2013, *arXiv:1312.4400*. [Online]. Available: http://arxiv.org/abs/1312.4400
[41] G. Tsoumakas and I. Katakis, "Multi-label classification: An overview," *Int. J. Data Warehousing Mining*, vol. 3, no. 3, pp. 1–13, 2007.
[42] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
[43] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015, *arXiv:1502.03167*. [Online]. Available: http://arxiv.org/abs/1502.03167
[44] B. Wang, X. Zheng, B. Qu, and X. Lu, "Retrieval topic recurrent memory network for remote sensing image captioning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 256–270, 2020.
[45] B. Wang, X. Lu, X. Zheng, and X. Li, "Semantic descriptions of high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 8, pp. 1274–1278, Aug. 2019.
[46] F. Zhang, B. Du, and L. Zhang, "Saliency-guided unsupervised feature learning for scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2175–2184, Apr. 2015.
[47] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th Int. Conf. Adv. Geographic Inf. Syst. (SIGSPATIAL GIS)*, 2010, pp. 270–279.
[48] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "BLEU: A method for automatic evaluation of machine translation," in *Proc. 40th Annu. Meeting Assoc. Comput. Linguistics (ACL)*, 2001, pp. 311–318.
[49] R. Vedantam, C. L. Zitnick, and D. Parikh, "CIDEr: Consensus-based image description evaluation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 4566–4575.
[50] C.-Y. Lin, "Rouge: A package for automatic evaluation of summaries," in *Text Summarization Branches Out*. Barcelona, Spain: Association for Computational Linguistics, 2004.
[51] M. Denkowski and A. Lavie, "Meteor universal: Language specific translation evaluation for any target language," in *Proc. 9th Workshop Stat. Mach. Transl.*, 2014, pp. 376–380.
[52] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.



Qi Wang (Senior Member, IEEE) received the B.E. degree in automation and the Ph.D. degree in pattern recognition and intelligent systems from the University of Science and Technology of China, Hefei, China, in 2005 and 2010, respectively.

He is a Professor with the School of Computer Science and the Center for OPTical IMagery Analysis and Learning, Northwestern Polytechnical University, Xi'an, China. His research interests include computer vision and pattern recognition.



Wei Huang (Student Member, IEEE) received the B.E. degree in control theory and engineering from Northwestern Polytechnical University, Xi'an, China, in 2018, where he is pursuing the M.S. degree in computer science with the Center for OPTical IMagery Analysis and Learning.

His research interests include deep learning and remote sensing.



Xueting Zhang received the B.E. degree in control theory and engineering from Northwestern Polytechnical University, Xi'an, China, in 2018, where she is pursuing the M.S. degree in computer science with the Center for OPTical IMagery Analysis and Learning.

Her research mainly focuses on remote sensing image processing.

Xuelong Li (Fellow, IEEE) is a Professor with the School of Computer Science and the Center for OPTical IMagery Analysis and Learning, Northwestern Polytechnical University, Xi'an, China.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:26:20 UTC from IEEE Xplore. Restrictions apply.